

RAHIEM JEROME BROOKS

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RESEARCH PROFILE

Interdisciplinary developmental researcher with training in developmental psychology, neuroscience, and applied intervention science. Designs and tests research-informed digital tools that support early childhood development and caregiver learning, and examines how caregiving itself is conceptualized and measured across cultural contexts — asking what dominant developmental frameworks, built largely on dyadic Minority World caregiving, fail to capture, and how distributed and participatory caregiving can be built into the design and measurement of family-facing interventions. Translates developmental research into accessible supports for families facing adversity. Combines quantitative analysis, intervention design, and science communication across research, policy, and practice settings. Applies the same measurement principles in applied settings, including the design and statistical validation of performance measurement frameworks in a corporate analytics role.

EDUCATION

University of North Carolina at Chapel Hill — M.A., Educational Innovation, Technology, and Entrepreneurship (MEITE)

Summer 2026 – Summer 2027 (Anticipated)

UNC School of Education

- Graduate coursework completed (Summer 2026): Introduction to Statistical Methods (EDUC 710) — High Pass (H); Quantitative Analysis for Public Policy (PLCY 460) — High Pass (H).

University of North Carolina at Chapel Hill — B.A., Psychology (with Neuroscience Coursework), Minor in History

May 2026

- GPA 3.58 · Graduated with Highest Honors and with Distinction · Carolina Research Scholar · Dean's List (Spring 2025, Spring 2026) · NC GlaxoSmithKline Scholar · Carolina Covenant Scholar

DIS Study Abroad — Copenhagen, Denmark

Summer 2025

UNC-accredited PSYC 575: Positive Psychology (3 credits)

Central Piedmont Community College — Associate of Arts

May 2024

- GPA 3.6 · Dean's List · C-STEP Scholar · Executive Vice President, Phi Theta Kappa Honor Society

CERTIFICATIONS AND ADDITIONAL TRAINING

Data Analytics and Visualization Certificate, UNC Charlotte School of Professional Studies

December 2023

- Six-month intensive (four evenings/week); 27 completed projects archived on GitHub. Skills: Python, R, SQL, JavaScript, machine learning, deep learning, ETL pipelines, geospatial visualization, NoSQL, API development.

Cornell Translational Research Summer Institute (CTRSI), Bronfenbrenner Center for Translational Research, Cornell University

June 2026

- Two-and-a-half-day residential training institute (Ithaca, NY) on conducting research in real-world settings, delivered by Bronfenbrenner Center faculty across diverse translational research backgrounds. Coursework format — training, discussion, and reflection on building research partnerships with community organizations, agencies, and service settings, and on translating empirical findings into practice and policy decisions.

Odum Institute for Research in Social Science, UNC Chapel Hill

Registered, Fall 2026

- Introduction to Multilevel Modeling; Structural Equation Modeling; Introduction to Survey Development; Introduction to Focus Groups.
- CITI Certified (May 2026, October 2024).

QUANTITATIVE AND METHODOLOGICAL TRAINING

Statistical modeling: Multiple regression; binary logistic regression with odds ratios, Wilson score intervals, and McFadden pseudo R^2 ; structural equation modeling and multilevel modeling (Odum Institute, Fall 2026).

Measurement and psychometrics: Composite scale construction; internal consistency (Cronbach's alpha, McDonald's omega); construct validity assessment across alternative weighting specifications; measurement validation; survey construction.

Nonparametric and effect-size methods: Spearman rank correlation, Wilcoxon signed-rank, Mann-Whitney U, effect-size estimation.

Simulation and inference: Monte Carlo simulation, sensitivity analysis, interval estimation, leave-one-out benchmarking.

Research design: Intervention design; four-phase primary data collection; IRB protocol development (CITI Certified); UX research and usability testing.

Open and reproducible science: Four OSF pre-registrations; reproducible analysis pipelines (R, Python); data codebooks; documented control totals for independent verification; method documentation to publication standard.

Statistical software: R, Python (pandas, matplotlib, seaborn), SQL, Stata, Jamovi, Qualtrics.

RESEARCH EXPERIENCE

Learning Science and Validation, CLICK Research Group

August 2026 – Present

UNC School of Education · PI: Dr. Jeffrey A. Greene, McMichael Professor of Educational Psychology and Learning Sciences

- Define and document the tagging constructs used to classify instructional content, specifying what each construct captures and where its boundaries sit.
- Identify the theoretical and empirical evidence grounding those constructs, drawing on the group's research in self-regulated learning and epistemic cognition.
- Develop the criteria against which AI-generated classifications are evaluated, and audit the validity of the resulting tags and derived measures.
- Ensure interpretations drawn from enriched data remain supported by the available evidence rather than by the volume or confidence of model output.

Graduate Research Assistant, Math Potentials Lab

May 2026 – Present

UNC School of Education · PI: Dr. Jessica Rodrigues · NSF CAREER Award #2237053 (Project KEY)

- Conducting graduate-level research on early mathematics learning and instructional design, contributing to data collection, coding, and analysis.
- Applying quantitative methods, intervention-design experience, and educational-technology training from the MEITE program to active lab projects.

Circle Lab

October 2024 – Present

UNC-Chapel Hill · PI: Dr. Margaret Sheridan

Graduate Research Assistant (May 2026 – Present)

- Supporting ongoing developmental neuroscience research on cognition, emotion regulation, and early adversity.
- Maintaining responsibilities for data management, participant confidentiality, and protocol fidelity (CITI Certified, October 2024).

Honors Researcher, The PANDA Study — PSYC 693–694H (Fall 2025 – Spring 2026)

- Honors thesis defended: “The PANDA Study: Feasibility of a Neuroscience-Informed Parenting Intervention for Early Childhood Development” (Spring 2026).
- Independently designed a neuroscience-informed parenting intervention integrating parent education and a child learning interface, delivered to caregivers of children ages 3–5.
- Developed the PANDA Learning Hub (HTML-based) as a direct-to-caregiver digital intervention platform for early learning support.
- Built the PANDA User Testing Platform in HTML to capture structured usability data across four study phases.
- Independently collected primary data across four phases with caregivers of children birth to five: Caregiver Readiness and Interest Survey, Pre-Intervention Assessment, Post-Intervention Assessment, and App Usability Testing.
- Documented caregiving barriers and family stress across the full birth-to-five range from the Waves 1–2 caregiver sample; these findings set the design targets for the intervention.
- Conducted quantitative analysis in R, including composite scale construction, multiple regression, Spearman correlations, Wilcoxon signed-rank tests, Mann-Whitney U tests, and effect-size interpretation.
- Applied psychometric reliability analysis including Cronbach's alpha and McDonald's omega; updated IRB protocols for family recruitment and participation.
- Received the David Bray Peele Memorial Research Award (\$250) to fund participant compensation; manuscript from thesis data submitted for publication.

Independent Researcher — PSYC 395 (Spring 2025 – Spring 2026)

- Conducted an IRB-approved study on neuroscience-informed parent engagement and early language development that laid the groundwork for the PANDA Study honors thesis.
- Designed and analyzed the PANDA Parent Survey in Qualtrics with attention to data security and participant confidentiality; created the study website (thepandastudy.com).
- Awarded the Summer Undergraduate Research Fellowship (SURF, \$4,000) to fund the independent project.

Research Assistant, The WHALE Study — Well-being, Health, and Adolescent Life Experiences (Oct 2024 – Spring 2025)

- Supported research on emotion regulation and cognitive development using behavioral and interview data; managed data cleaning, verification, and confidentiality protocols (CITI Certified).

- Applied developmental neuroscience insights from lab work directly to the design of the PANDA Study intervention.

Applied Research and Design Projects, UNC School of Education

Fall 2025 – Spring 2026

- EDUC 584: AI and Learning Innovation (Fall 2025) — applied AI frameworks to early-learning intervention design.
- Designed the PANDA Bridge in EDUC 761 (Design of Emerging Technologies for Education) — a professional-facing decision-support tool for pediatricians, social workers, and home-visiting providers to translate developmental science into family-feasible guidance at the point of clinical interaction.
- Conducted original quantitative UX research for Kreebo, a new educational application, in EDUC 590 (UX Research in Education), producing a full findings report presented directly to the developer.

Finalist and Participant, Duke/UNC Advancing Diversity in Aging Research (ADAR)

Fall 2024 – Spring 2025

- Trained in cognitive aging, resilience, and lifespan developmental frameworks through equity-focused mentorship; applied lifespan theory to connect early- and later-life developmental interventions.

MANUSCRIPTS UNDER REVIEW AND IN PREPARATION

- Brooks, R. J. (2026). Engagement patterns and caregiver-reported outcomes in a brief digital parenting intervention for a socioeconomically diverse sample of caregivers: A feasibility pilot [Manuscript submitted for publication]. *PLOS ONE*.
- Brooks, R. J., & Adimora, D. E. (2026). *Translating Nigerian family practices to ease caregiving barriers for low-income U.S. families* [Manuscript submitted for publication]. *Translational Issues in Psychological Science*. Critical review proposing a seven-domain typology of documented Nigerian family practices and a translational method separating developmental process from cultural expression. Contribution: first author; conceptual framework, typology development, and translation-readiness analysis.
- Brooks, R. J., Ibekwe-Okafor, N., & Enelamah, N. V. (in preparation). Whose caregiving counts? A critical reanalysis of early childhood development indicators in MICS Nigeria and Ghana. Pre-registered on the Open Science Framework (<https://osf.io/ds36w>). Letter of Intent submitted to *Child Development*, Special Section on "Centering the Majority World in Developmental Science" (July 2026). Anticipated full manuscript submission: December 2026.
- Brooks, R. J. (2026). *Participatory caregiving repertoires and early development in the Majority World: A mixed-methods study of Ethiopia and Peru using Young Lives data* [Letter of intent submitted for the special section "Centering the Majority World in Developmental Science," *Child Development*]. Pre-registered on the Open Science Framework (<https://osf.io/gzcgqr>). Anticipated full manuscript submission: December 2026.
- Ibekwe-Okafor, N., Enelamah, N. V., Kalu, C., Brooks, R. J., & Ngwayarudza, P. (2026). *Rethinking family assets in early childhood: Understanding children's development through collective family contributions* [Letter of intent submitted for the special section "Centering the Majority World in Developmental Science," *Child Development*]. Advances the Distributed Family Assets Framework using structural equation modeling with the 2021 Nigeria Multiple Indicator Cluster Survey (MICS6; $N \approx 8,000$ children ages 3–4). Contribution: measurement and analysis.
- Rodrigues, J., Singell, E. L., Dash, S., & Brooks, R. J. (in preparation). Communicating research-informed mathematics recommendations to teachers: A systematic analysis of practitioner journal articles. Math Potentials Lab, UNC–Chapel Hill; NSF CAREER Award #2237053 (PI: Dr. Jessica Rodrigues). Pre-registered on the Open Science Framework (OSF Registries, July 10, 2026). Anticipated submission: August 2026. Contribution: authored the manuscript introduction and literature review.
- Brooks, R. J., & Rodrigues, J. (2026). *Two kids get the same answer. Only one understands fractions* [Manuscript completed; submission pending]. *Frontiers for Young Minds*. Core Concept article (Mathematics and Economics section) translating research on fraction magnitude, representational translation, and deep learning for readers ages 8–11; argues that answer-only assessment cannot distinguish a memorized procedure from conceptual understanding. Contribution: first author; conceptualization, writing, and figure design.

PRESENTATIONS

- Brooks, R. J. (2026). The PANDA Study. Poster accepted for presentation, 59th Annual Meeting of the International Society for Developmental Psychobiology, New Orleans, LA.
- Brooks, R. J. (2026). The PANDA Study. Poster, Celebration of Undergraduate Research, UNC–Chapel Hill.
- Brooks, R. J. (2026). The PANDA Study. Annual Meeting of the Society for Neuroscience (SfN).
- Rodrigues, J., Gustaveson, A., Duncanson, E., Yu, B., Mukashyaka, A., Schulze, S., Brooks, R. J., Dos Santos Araujo Lourenco, A., Scotti D'abbusco, M., & Aghoram, A. (2026, September). Unlocking math success: Disseminating research-backed recommendations to support students experiencing math difficulty. Poster presented at the North Carolina Annual Inclusion Summit, Appalachian State University, Boone, NC.
- Brooks, R. J. (2025). Maximizing early brain development in underserved children. Poster, Scholarship and Social Justice Undergraduate Research Conference, Harvard University.
- Brooks, R. J. (2025). Maximizing early brain development in underserved children. Poster, Celebration of Undergraduate Research, UNC–Chapel Hill.

- Brooks, R. J. (2025). Maximizing early brain development in underserved children. Poster, National Psychology Summit, Psi Beta National Honor Society (Virtual).

PRE-REGISTRATIONS

- Brooks, R. J. (2026, July 30). *Participatory caregiving repertoires and early development in the Majority World: A mixed-methods study of Ethiopia and Peru using Young Lives data* [Preregistration]. OSF Registries. <https://osf.io/gzccq> — Secondary Data Preregistration (confirmatory quantitative analyses); currently embargoed. Letter of Intent submitted to *Child Development*, Special Section on “Centering the Majority World in Developmental Science” (July 2026).
- Brooks, R. J., Ibekwe-Okafor, N., & Enelamah, N. V. (2026, July 10). *Whose caregiving counts? A critical reanalysis of early childhood development indicators in MICS Nigeria and Ghana* [Preregistration]. OSF Registries. <https://osf.io/ds36w> — Currently embargoed; open data, analytic code, and materials to be released on publication. Letter of Intent submitted to *Child Development*, Special Section on “Centering the Majority World in Developmental Science” (July 2026); anticipated full manuscript submission December 2026.
- Rodrigues, J., Singell, E. L., Dash, S., & Brooks, R. J. (2026, July 10). *Phase 1: Project KEY systematic journal analysis* [Systematic review registration]. OSF Registries. Generalized Systematic Review Registration, Math Potentials Lab, UNC–Chapel Hill; NSF CAREER Award #2237053 (PI: Dr. Jessica Rodrigues).
- Dash, S., Rodrigues, J., Singell, E. L., & Brooks, R. J. (2026, July 4). *Knowledge exchange for supporting youth with or at risk for mathematics difficulty* [Preregistration]. OSF Registries. Math Potentials Lab, UNC–Chapel Hill; NSF CAREER Award #2237053 (PI: Dr. Jessica Rodrigues).

POLICY TRANSLATION AND RESEARCH PRODUCTS

Policy Briefs (Society for Research in Child Development)

- Brooks, R. J. (2026, May). Maternal employment, paid leave, and childcare: Comprehensive evidence extraction for the Office of Senator Cantwell. Memorandum prepared for the Office of U.S. Senator Maria Cantwell (D-WA), Ranking Member, Senate Committee on Commerce, Science, and Transportation, via SRCD; reviewed by SRCD and delivered to congressional staff following a briefing with the Senator's office.
- Brooks, R. J. (2026). The impacts of juvenile detention and delayed mental health care on youth experiencing mental health crises: Evidence summary for policy consideration. Policy brief prepared through SRCD.

PANDA Study — Research Products Developed

- PANDA Learning Hub (HTML direct-to-caregiver intervention platform); PANDA User Testing Platform (HTML usability-data capture); PANDA Bridge (professional-facing decision-support tool).
- Caregiver Readiness and Interest Survey, Pre-/Post-Intervention Assessments, and App Usability Testing Protocol (Qualtrics + caregiver-facing instruments).
- Data codebook and reproducible R analysis pipeline; thepandastudy.com public research website.

CONFERENCE PARTICIPATION

ACF National Research Conference on Early Childhood (NRCEC 2026)

June 22–24, 2026

- Administration for Children and Families, Office of Planning, Research, and Evaluation · Crystal Gateway Marriott, Arlington, VA. Attended as a graduate researcher in early childhood development. National convening of researchers, program administrators, and federal policymakers, held annually since 1991; program spanned plenary sessions, symposia, poster symposia, and professional development sessions on Head Start, child care, home visiting, child welfare, pre-kindergarten, and early elementary research and its applications to policy and practice.
- Participant, Emerging Scholars Research and Mentoring Event (Session 209; June 23, 2026; pre-registration required) — structured mentoring program matching emerging scholars with senior early childhood researchers by research interest; paired with a faculty mentor for discussion of research direction, doctoral pathways, and career development in early childhood research.

PROFESSIONAL EXPERIENCE

Senior Quantitative Analyst, Receivables Analytics, SOMPO International

March 2026 – Present

Global specialty insurance and reinsurance (28+ countries) · Full-time remote

- Designed and validated the performance measurement framework for a 16-analyst department, covering a \$1.04B receivable across 1,246 broker accounts, 21 portfolios, and 63,932 transaction records.
- Applied binary logistic regression at the observation level ($n = 3,710$; $n = 2,490$) to isolate individual effects from case-mix confounds, reporting odds ratios, Wilson score confidence intervals, and McFadden pseudo R-squared with explicit interpretation of unexplained variance.
- Developed a leave-one-out peer benchmarking procedure to address circularity in performance comparison, reporting coverage for each estimate so that low-coverage comparisons are identified as indicative rather than conclusive.

- Conducted Monte Carlo simulation (4,000 iterations) to test a proposed allocation rule against its stated objective, demonstrating that the rule could not achieve equalization under any random configuration.
- Assessed construct validity of a composite performance index across eight weighting specifications; identified rank instability of up to ten positions and recommended against publication as an ordinal measure.
- Evaluated five candidate operationalizations of a proposed departmental target, demonstrating that measured performance ranged from 59.7% to 97.2% depending on the operational definition selected.
- Identified systematic misclassification affecting 2,009 records and \$32.0M in reported balances; recalculated all downstream estimates following correction.
- Authored a methods glossary documenting every statistical procedure applied, with definitions, applications, and reliability justifications referenced to published sources (APA 7th ed.).

Cash Analyst, SOMPO International

June 2023 – March 2026

Global specialty insurance and reinsurance (28+ countries) · Full-time remote

- Managed insurance policy reconciliation with outstanding receivables of \$126M using Python, Excel, and Tableau.
- Applied pandas, matplotlib, and seaborn to extract insights and build interactive visualizations; developed finance dashboards for aging-bucket issues and balance tracking by broker and insured account.
- Collaborated with finance teams and a third-party collection agency to present findings and manage policy placement, tracking, and reporting.

Marketing Manager, Prodigy Gold Books

September 2017 – April 2023

Independent publishing company · Philadelphia, PA and Charlotte, NC

- Built marketing-mix models and visualizations using SQL, Python, and Tableau (8 bp ROI lift); identified opportunities saving \$11,000+ in annual campaign costs.
- Managed a team of five; streamlined processes (75% paperwork reduction); analyzed trends yielding a 10% sales-revenue increase and 15% cost reduction.

Marketing Associate, Prodigy Gold Books

March 2016 – September 2017

- Managed press social-media accounts, served as national media contact, and assembled print and electronic press materials.

Accounts Payable and Receivable Associate, Prodigy Gold Books

November 2006 – March 2016

- Managed high-volume daily invoices (up to a quarter-million dollars) using SAP S/4HANA and Excel; prepared financial reports, reconciled statements, and resolved discrepancies.

TECHNICAL SKILLS

Programming & Analysis: Python (pandas, matplotlib, seaborn), R, SQL, SQLAlchemy, JavaScript, HTML, VBA.

Data Tools & Platforms: Jupyter Notebooks, Tableau, Excel (Advanced), Jamovi, Qualtrics, Stata.

Data Engineering: NoSQL, Apache SparkSQL, ETL pipelines, API development, web scraping, geospatial visualization (Leaflet), machine learning, deep learning.

Application Development: Built the PANDA Learning Hub, PANDA Bridge, and PANDA User Testing Platform independently in HTML; created thepandastudy.com.

PUBLICATIONS (NON-ACADEMIC)

Author of 12 published works (10 fiction, 2 nonfiction), 2009–2018 (available on [Goodreads](#)). Experience in long-form writing, narrative construction, and translating complex human experiences into accessible prose for general audiences.

LEADERSHIP AND MEMBERSHIPS

- Executive Board Member, Community Events — Carolina Neuroscience Club (Aug 2025 – May 2026): grew event attendance 30% through community partnerships.
- Treasurer and Inaugural Executive Board Member — Black Minds in Neuro (Mar 2025 – May 2026): managed budget; secured \$800 in funding via UNC Senate proposals.
- Mentorship Committee Member (Black on Track) — Union of Black Men (Sept 2024 – Present): mentored first-year students through academic transitions.
- Fellow, Summer Research–Early Identification Program (SR-EIP) — The Leadership Alliance, Brown University (Summer 2024 – Present).
- McNair Scholars Program (Fall 2024); SEOPP Survivor Scholarship, McNair Promising Practices Institute.
- Member: SRCD (2026–); Psi Chi (2025–); Carolina Neuroscience Club; Phi Alpha Theta; Alpha Alpha Alpha (First-Generation Honor Society).

HONORS AND AWARDS

- Summer Undergraduate Research Fellowship (SURF), \$4,000.
- J. Steven Reznick Diversity and Psychological Research Grant, Honorable Mention (2026).
- David Bray Peele Memorial Research Award, \$250 — competitive departmental grant, UNC Department of Psychology and Neuroscience.
- UNC C-STEP Pathways to STEM Success Summer Immersion Scholarship (\$3,250); SEOPP Survivor Scholarship (\$500).
- Olmstead Family Abroad Scholarship (\$2,000); Fund for Education Abroad Scholarship (\$1,250); Gilman Scholarship (\$2,500).
- NC GlaxoSmithKline Scholar · Carolina Covenant Scholar.

RELEVANT INTERDISCIPLINARY COURSEWORK

Policy & Quantitative Analysis: Quantitative Analysis for Public Policy (PLCY 460, High Pass, Summer 2026).

Quantitative & Research Methods: Statistical Principles of Psychological Research; Introduction to Data Models and Inference; Research Methods in Psychology; Honors Psychology (PSYC 693–694H); Independent Research (PSYC 395); Introduction to Statistical Methods (EDUC 710, High Pass, Summer 2026).

Maternal, Child & Developmental Health: Child Development; Human Growth and Development; Clinical Psychology.

Educational Technology & Applied Innovation: AI and Learning Innovation (EDUC 584); Innovation in Teaching (EDUC 614); Design of Emerging Technologies for Education (EDUC 761); UX Research in Education (EDUC 590).